

RETROPIE

GAMING CONSOLE ON RASPBERRY PI

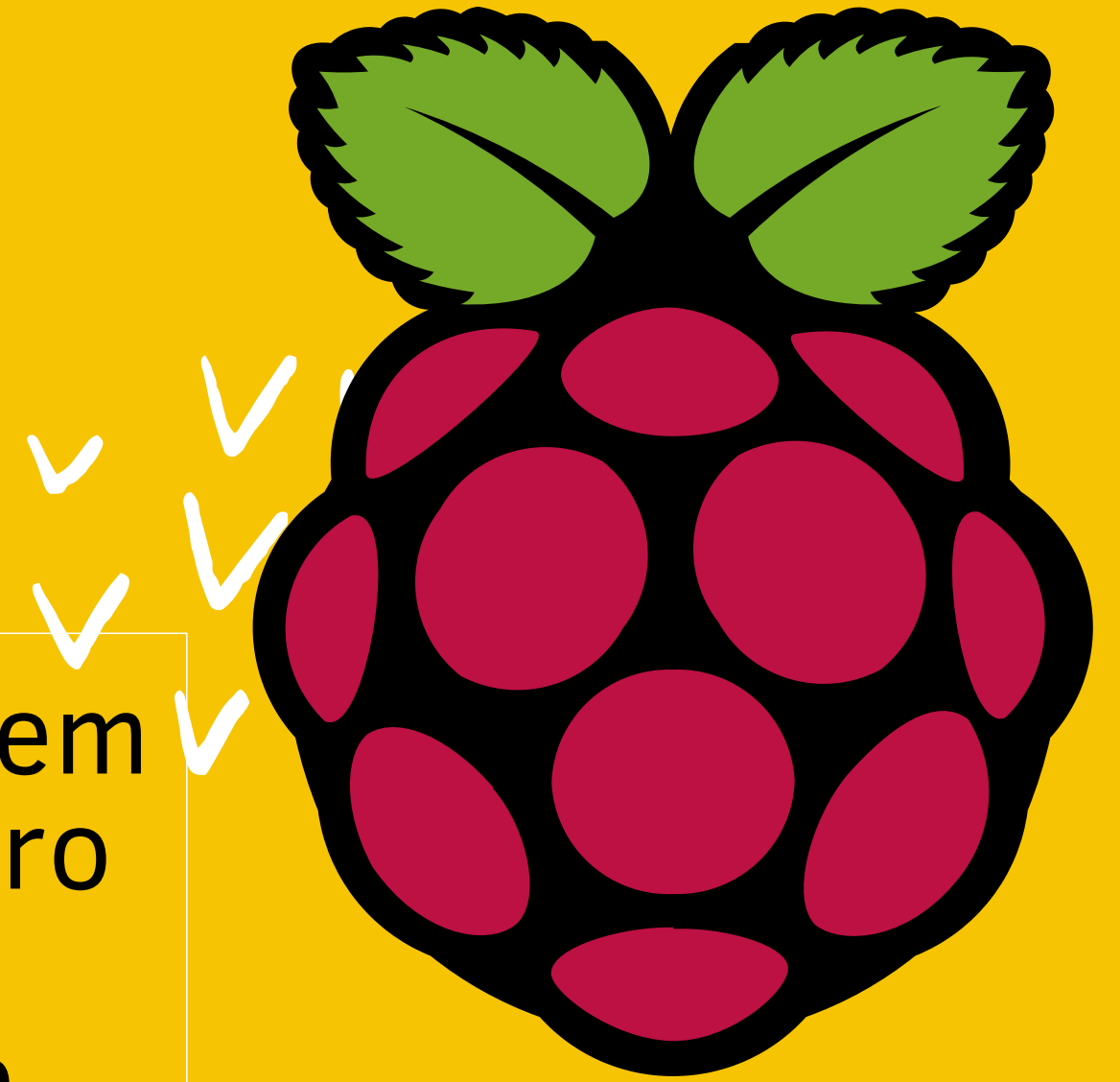
Operating Systems - Lab

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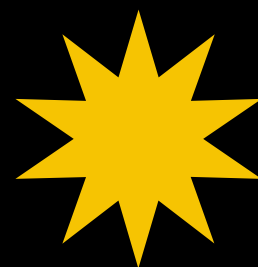


WHAT IS RETROPIE?

RetroPie is a Linux-based operating system that transforms a Raspberry Pi into a retro gaming console by combining emulation software, system services, and hardware abstraction layers.

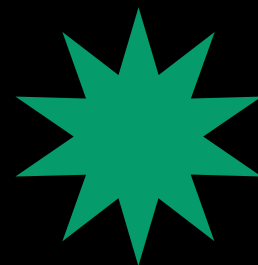


WHY THIS PROJECT?



Problem:

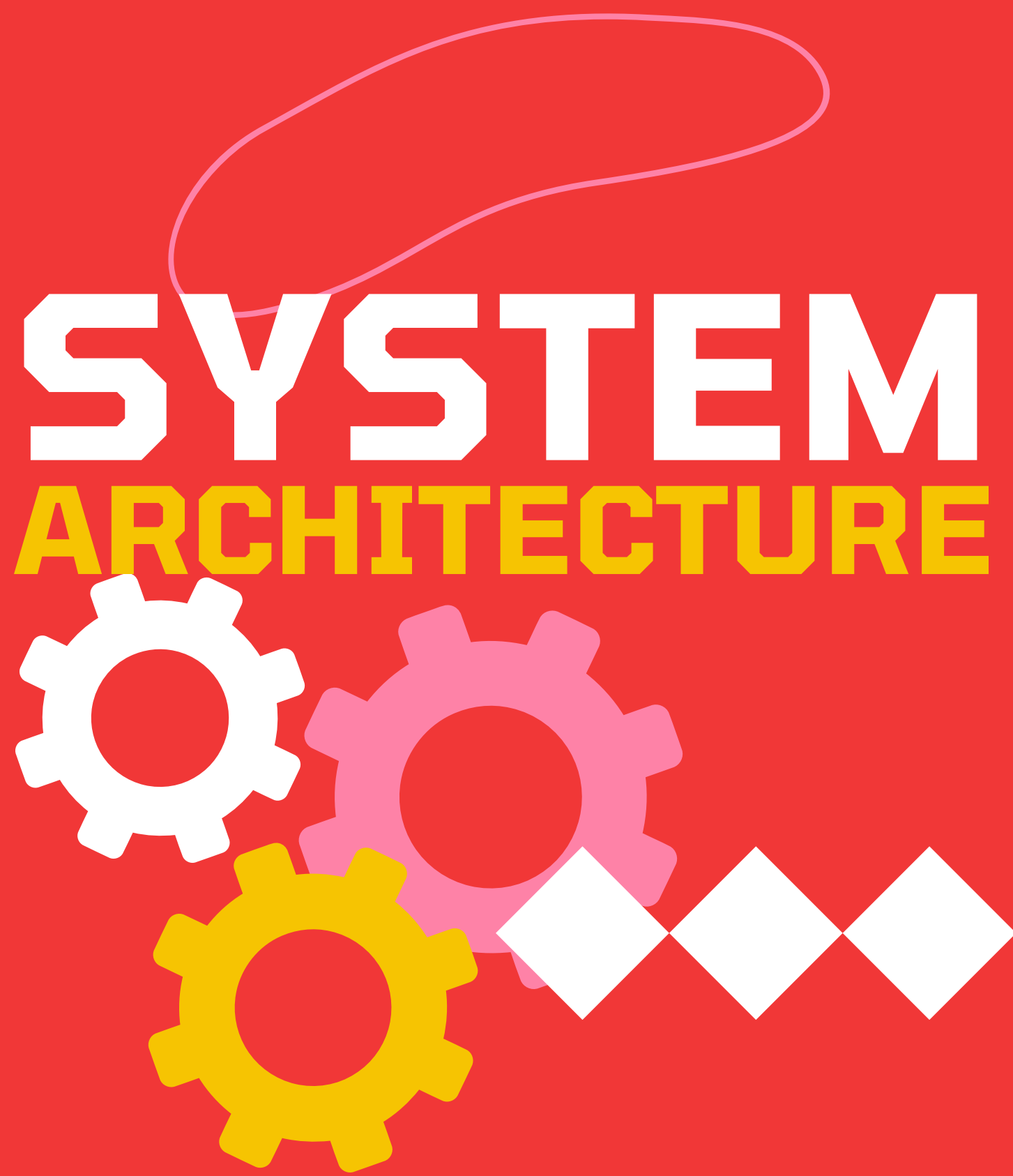
- Complex OS concepts are abstract
- Hard to visualize hardware-software interaction



Solution:

- RetroPie provides a real OS
- Runs on actual hardware
- Shows boot, processes, memory, I/O in action.





SYSTEM ARCHITECTURE

User Interface

→ EmulationStation

Applications

→ RetroArch & Emulators

System Services

→ systemd, SSH, Samba

Operating System

→ Raspberry Pi OS

Kernel

→ Linux (ARM)

Hardware

→ Raspberry Pi 4

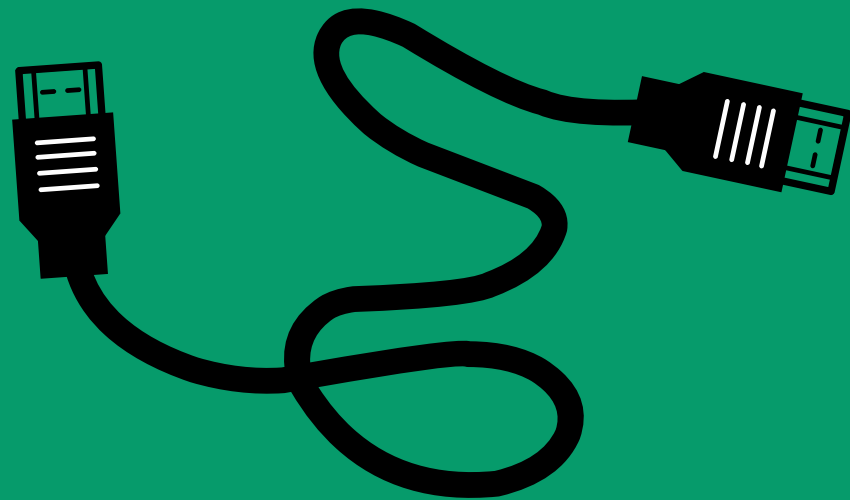
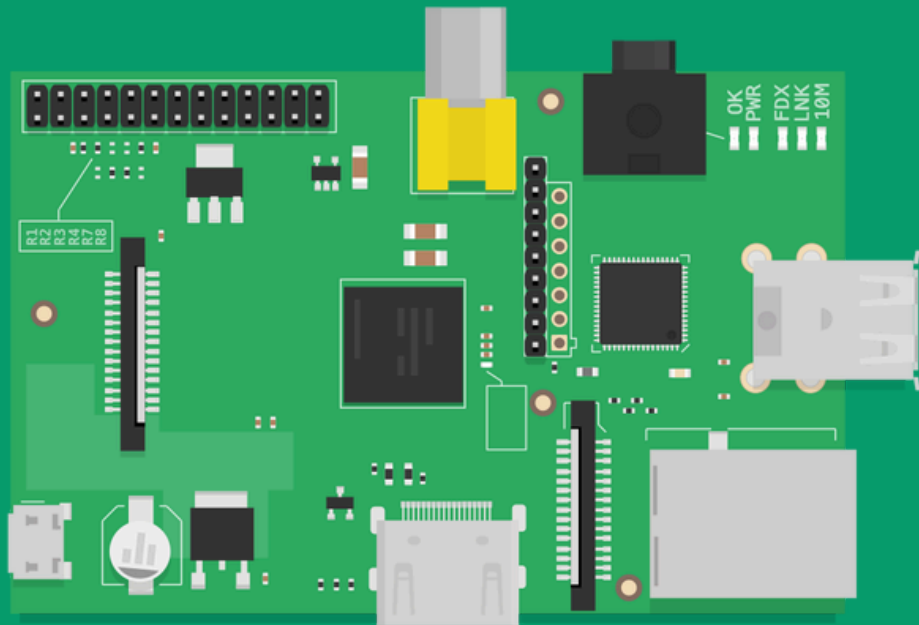
OS CONCEPTS IN ACTION

OS Concept	Where it Appears
Process Management	Emulator processes
Memory Management	RAM & GPU split
I/O Management	Controller + HDMI
File System	ROM directories
Boot Process	Auto-start EmulationStation

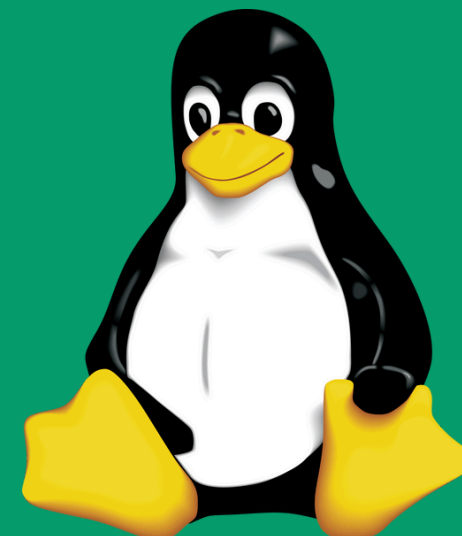
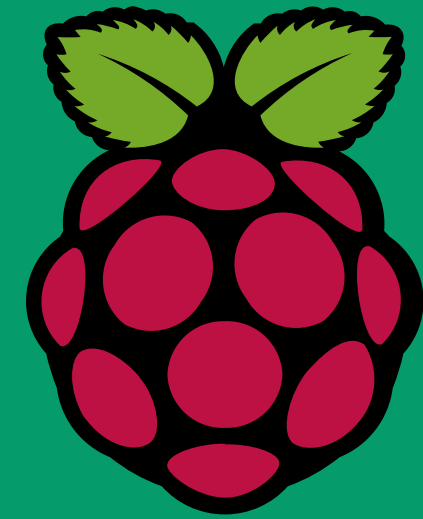


HARDWARE & SOFTWARE STACK

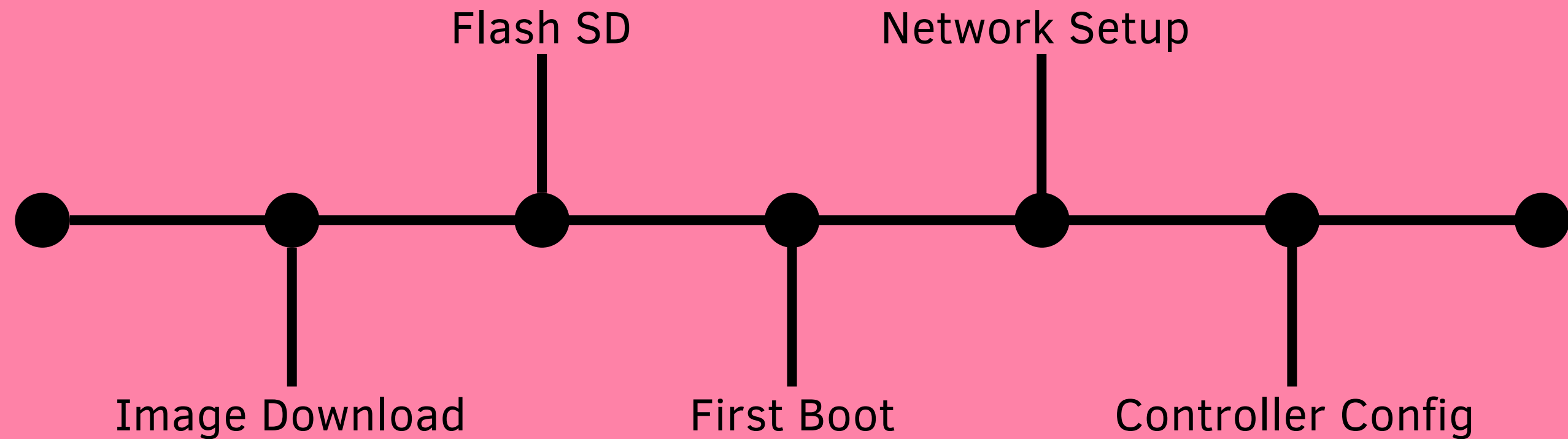
Hardware



Software



INSTALLATION JOURNEY



ROM & FILE SYSTEM DESIGN



/home/pi/RetroPie/roms/

nes/

mario.nes

zelda.nes

snes/

metroid.smc

donkey_kong.smc

psx/

crash_bandicoot.bin

final_fantasy.img

REAL CHALLENGE FACED

A low-quality microSD card caused OS flashing failures, screen flickering, and unstable gameplay.

ROOT CAUSE ANALYSIS



**LOW-QUALITY
SD CARD**

Unstable
Read/Write
Speeds

**DISPLAY FLICKERING
& BOOT ISSUES**



WHAT WE LEARNED

This project transformed theoretical operating system concepts into observable system behavior, especially in areas of boot processes, memory allocation, and I/O handling on embedded hardware.



CONCLUSION

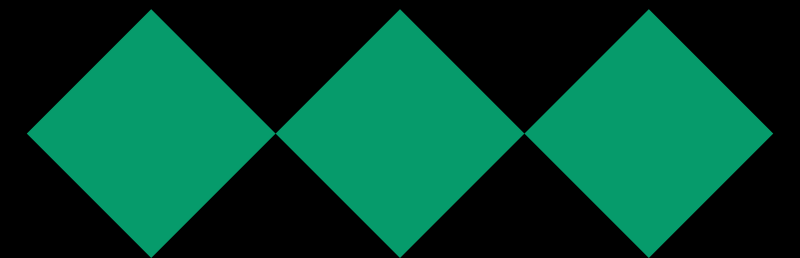


Successfully deployed a Linux OS

Demonstrated real OS internals

Linked software behavior with
hardware constraints

Gained embedded systems experience



The background is a solid black field. It features several geometric elements: a green square in the top-left corner, a pink square in the top-right corner, and a blue cloud-like shape at the bottom center. A series of thin, dark grey lines radiate from the center of the image towards the corners. There are also several white and red line art elements: a large white wavy line on the left, a red curved line with multiple parallel strokes on the upper left, and a white scribbled line on the upper right. In the bottom right corner, there are red curved shapes resembling stylized waves or parts of a larger design.

THANK
YOU